La Scuola d'Italia / Dr. Huson / IB Math 11
3 November 2026

Nickname:

2.8 Classwork: Systems of Equations by Elimination

1. Solve each system of equations using the *elimination* method. Show the equation you get after eliminating one variable. You may check your answers with your GDC.

(a) $3x + 2y = 17$
$5x - 2y = 7$

(b) $5x + 3y = 21$
$4x - y = 27$

(c) $3x + 4y = 14$
$2x + 3y = 11$

(a)

(b)

(c)

2. A city studies the weekly market for bicycle helmets. The quantity is q hundred helmets per week and the price is p euros per helmet. Two surveys give these results.

	Demand (shoppers buy)		**Supply** (shops offer)	
Price p (euros)	36	72	33	68
Quantity q (hundreds)	16	4	6	20

Demand and supply are each modelled by a straight line $p = mq + c$.

(a) Show that the demand line is $p = 84 - 3q$.

(b) Find the equation of the supply line.

(c) Write both equations in the form $aq + bp = c$. Hence, use the elimination method to find the equilibrium quantity and price.

(d) The city sets a minimum price of 60 euros per helmet. Find the quantity demanded and the quantity supplied at this price. Hence, find the surplus of helmets each week.